Inorganic Cumulative Exam
Oct 17 2025

Answer any 5 of the following 10 questions (20 points each). Any answers provided beyond the requested five will not be graded. Remember: Your answers to these questions are meant to provide evidence of broad, graduate-school level chemistry knowledge. Consider this in your choice of terminology and examples as answers to these questions.

1. The existence of high-spin octahedral complexes is an apparent violation of the Aufbau principle. Use the relevant frontier molecular orbital diagrams to answer the following questions:
   (a) (10 pts) What is the Aufbau principle and why does the high spin octahedral complex appear to violate it? Use a d^7 high spin octahedral complex spin as an example .
   (c) (10 pts) Why are tetrahedral complexes rarely (if ever) found in a high-spin configuration?
2. (a) (10 pts) Define hydration isomerism.
   (b) (10 pts) Give an example of two compounds that are hydration isomers.

3. (a) (5 pts) How many vibrational modes does molecular H_2O have? Justify your answer.
   (b) (15 pts) Sketch each vibrational mode of H_2O, indicating motions with arrows and/or description as needed.
4. Define the photoelectric effect and describe an experiment that allows it to be used to identify the absolute orbital energies of elements used in building molecular orbital diagrams?
5. Use the included table of standard reduction potentials to answer the following questions
   (a) (10 pts) A Zn metal electrode is placed in an aqueous solution of $CuSO_4$. What happens? Explain your answer in terms of the half reactions of interest, the given standard reduction potentials, and the Gibbs free energy of the system.

Reduction half-equation	E^o / V
$Ca^{2+}(aq) + 2e^- \rightleftharpoons Ca(s)$	−2.87
$Ti^{2+}(aq) + 2e^- \rightleftharpoons Ti(s)$	−1.63
$V^{2+}(aq) + 2e^- \rightleftharpoons V(s)$	−1.18
$Cr^{2+}(aq) + 2e^- \rightleftharpoons Cr(s)$	−0.91
$Mn^{2+}(aq) + 2e^- \rightleftharpoons Mn(s)$	−1.19
$Fe^{2+}(aq) + 2e^- \rightleftharpoons Fe(s)$	−0.44
$Co^{2+}(aq) + 2e^- \rightleftharpoons Co(s)$	−0.28
$Ni^{2+}(aq) + 2e^- \rightleftharpoons Ni(s)$	−0.25
$Cu^{2+}(aq) + 2e^- \rightleftharpoons Cu(s)$	+0.34
$Zn^{2+}(aq) + 2e^- \rightleftharpoons Zn(s)$	−0.76

   (b) (10 pts) A Zn metal electrode is placed in an aqueous solution of $MnSO_4$. What happens? Explain your answer in terms of the given standard reduction potentials and Gibbs free energy.

(Figure for question #6)

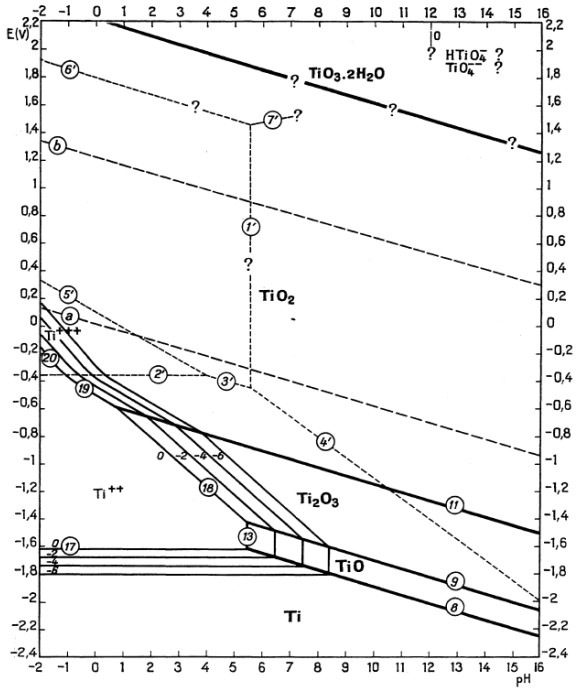


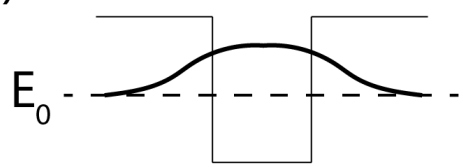
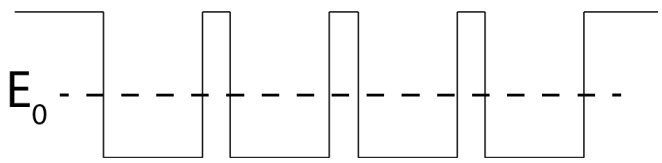
FIG. 1. Potential–pH equilibrium diagram for the system titanium–water, at 25°C. [Figure established by considering, as derivatives of tri- and tetravalent titanium, the anhydrous oxides Ti_2O_3 and TiO_2 (rutile).]

6. For the following questions, refer to the Pourbaix diagram for a dilute aqueous solution of Titanium (above). The y-axis (Potential [V]) is measured with respect to the Normal Hydrogen Electrode (NHE) potential.
    a. (5pts) What do the dashed lines (a) and (b) represent?
    b. (5 pts) Under what potentials and pH is the *divalent* titanium oxide thermodynamically stable?
    c. (5 pts) Under what conditions can a metallic layer of Ti be formed from the solution?
    d. (5 pts) Describe any chemical transformations that will occur on a metallic Ti foil when using it for water oxidation.
7. (a) (15 pts) Given a specific compound (Fe_2O_3, for example) describe the similarities and differences that an x-ray diffraction (XRD) pattern of a nanopowder (particle size < 10nm) of the compound different than the x-ray diffraction pattern of a powder comprised of microcrystals (particle size > 1 µm)?
   (b) (5 pts) How can one determine the characteristic grain size of a sample from the XRD pattern?
8. (a) (10 pts) Define the term 'exciton', an important quasiparticle in light-matter interactions.
   (b) (5 pts) What is meant by the term, "an exciton's binding energy"?
   (c) (5 pts) Why is the term exciton used more often when describing polymer and molecular systems in contrast to semiconductors?
9. (a) (12 pts) Explain a van der Waals attraction in terms of electrostatics.
   (b) (8 pts) Is this different than ionic bonding? Why or why not?

10. A simple derivation of band theory in solids or polymers involves building increasingly longer one-dimensional chains of identical potential wells. In the figure below (a) shows the ground state, s-like electronic wavefunction of a single finite square well potential and its ground state energy E_0.
(a) (10 pts) Using this graphical result, draw the unique 'linear combinations of atomic orbitals' that represent the molecular orbitals for the four finite-square well chain
(b). (10 pts) Given your solutions rank their corresponding energies from the ground state to the highest energy solution.

(Figure for question #10)

(a)



(b)